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Unravelling the structure-performance relationship over iron-based Fischer-Tropsch synthesis by depositing the iron carbonyl in syngas on SiO_2 in a fixed-bed reactor

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ABSTRACT

A new process, *in situ* deposition method, was developed to unveil Fe phase evolution behavior in iron-based Fischer-Tropsch synthesis (FTS). In detail, Fe(CO)*x*-containing syngas (CO/H_2) was fed into a fixed-bed reactor where Fe(CO)*x* directly decomposes to metallic Fe with subsequent favorable carburization to form active χ-Fe_5C_2 on SiO_2 under FTS reaction conditions. This facile process gives a high χ-Fe_5C_2 content (80–90%) without Fe_3O_4 phase in spent catalyst. In contrast, the 5.3Fe/SiO_2 prepared by the conventional impregnation method results in very low χ-Fe_5C_2 content (14.5%) and high Fe_3O_4 content (47%) in spent catalyst, which is ascribed to the strong Fe oxide-SiO_2 interaction hindering its deep reduction into metallic Fe but terminating as FeO. The reduced FeO could not be efficiently carburized into χ-Fe_5C_2 and tends to oxidize into Fe_3O_4 during the FTS reaction. Furthermore, the factors determining CO_2 selectivity are systematically investigated based on abundant experiments. Results show that CO_2 selectivity is closely related to CO conversion, Fe phase composition, reaction conditions, and the presence of K promoter. In brief, when the factor enhances H*/C* ratio on χ-Fe_5C_2, the reaction of dissociative O* from CO shifts towards hydrogenation to H_2O despite that its reaction with CO* to form CO_2 *via* Boudouard mechanism is kinetically favorable. In addition, CO_2 selectivity could be reduced *via* reverse water-gas shift (RWGS) reaction on Fe_3O_4 with H_2O generation.

1. Introduction

Fischer-Tropsch synthesis (FTS) is a heterogeneous catalyzed polymerization reaction where the natural gas-, coal-, or biomass-derived syngas (a mixture of CO and H_2) is converted into a wide spectrum of hydrocarbon chains [1,2]. Most group VIII transition metals are active in the reaction, but only Fe and Co catalysts are typically used industrially [3–7], owing to their high selectivity to the desired long hydrocarbon chains and economic feasibility of the overall FTS process. Compared to the Co-based FTS catalysts, Fe-based catalysts have attracted more attention recently due to some merits, such as the low cost and wide availability of the metal, an easily tunable selectivity towards olefins and alcohols as the most important chemical raw materials, and excellent endurance in operation with regard to changing reaction conditions and poisoning [8–13]. Meanwhile, the Fe-based FTS reaction can also generate large amounts of CO_2 at relatively high temperature, which is generally considered from the water-gas shift (WGS) reaction [14,15]. When the H_2 rich syngas is applied to Fe-based FTS reaction or the H_2 supplying to the H_2 lean syngas originating from coal and biomass could be available by splitting H_2O by electro- or photo-catalysis [16–18], the *in-situ* re-adjustment of the H_2/CO molar ratio *via* WGS reaction is not necessary. Therefore, it is most important for us to develop a strategy to suppress the formation of CO_2 in Fe-based FTS reaction, which could improve the utilization efficiency of carbon and reduce the emission of the green-house gas [19,20].

Under FTS reaction conditions, the bulk Fe catalysts display a poor mechanical stability. Especially, Fe-based FTS reaction is often industrially operated in slurry bubble column reactors (SBCRs) or in fluidized bed reactors, which requires the catalysts to have an excellent resistance to attrition and to avoid the catalyst fragmentation resulting in operational problems. An effective strategy is to impregnate the active Fe component with those supports which have excellent mechanical stability or to use binder making the precipitated or hydrothermal synthesized Fe catalysts attrition-resistant. Although this strategy could improve the mechanical stability, the support or binder might result in negative or positive effects on FTS reaction performance, *e.g.* the catalytic activity and product selectivity. Moreover, to enhance the catalytic performance, the structural and electronic promoters, such as Bi

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[21], Pb [21], Mn [22,23], Ag [23], K [24], Na [22,25,26] and S [24–26], are often added into the Fe catalysts, which could overcome the deactivation owing to sintering, carbon deposition and iron phase evolutions, and regulate the selectivity to the desired product fraction. However, the promoter affects the reaction performance with complexity during the FTS reaction. For example, Mn could form the mixed oxide with Fe to hinder the reduction of Fe oxides and [23], however, the relatively easier reduction of MnO*x* could make the oxygen vacancy facilitate the reduction of FeO phase [22]. Also, MnO*x* serving as support might easily lead to the remarkable decline in activity due to the structural collapse resulting from easier reduction [23]. In the case of S promoter, on the one hand, it can suppress the chain growth to enhance the selectivity to light α-olefins [24,25]; on the other hand, it can also influence the reduction of Fe oxides and subsequent carburization into active χ-Fe_5C_2 phase [24]. In addition, for the conventional catalyst preparation and the FTS reaction process, Fe phase evolution starts from Fe oxides and forms the desired χ-Fe_5C_2 phase [27,28]. Due to complex changes of Fe phases during the calcination treatment, reduction treatment, carburization behaviors in different gas atmosphere, it brings about the difficulty in clearly understanding the evolution of Fe phases, and the effects of Fe phases and their nano-structures on reaction performance [29–38].

Based on the above concerns in Fe-based FTS catalysts, it is crucial to develop the facile methods for catalyst preparation and reaction process which could avoid the interplaying and confusing of the involved various parameters in affecting the mechanistic insights and in determining the catalytic performances. Only investigating by this way, it could provide more reliable and convincing elucidation over the complicated FTS reaction process. In our previously investigated Co-based catalyst system, it was found that the α-olefins could successively de-methylene to shorten the chain length to form lower *n*-hydrocarbons with methylene (:CH_2), and the :CH_2 could again insert into the alkyl-Co bond forming *via* the adsorption of the original α-olefins to form higher *n*-hydrocarbons or be directly hydrogenated to CH_4, which was observed by introducing the 1-dodecene with H_2 rather than syngas (CO/H_2) over supported 20% Co/SiO_2 catalyst in the fixed-bed reactor [39]. Also, we designed the reduction-oxidation-reduction (ROR) pretreatment method over 20% Co/SiO_2 catalyst to clearly unveil the evolutions of SiO_2 structure, Co phase and its morphology by simply replacing the oxidizing gas O_2 with H_2O in oxidation step [40], which provides fundamental insights into how to fabricate Co-based FTS catalysts and select operating conditions to obtain the advanced FTS catalysts and reaction performance.

In this study, over Fe-based FTS catalyst, a new process was designed that the catalyst synthesis and FTS reaction proceed simultaneously in a fixed-bed reactor. Briefly, SiO_2 support instead of Fe catalyst was firstly loaded in reactor. The iron carbonyl-containing syngas was then introduced under the FTS reaction conditions. By this way, the iron carbonyl could directly decompose into metallic Fe on SiO_2 in reductive environment (syngas), which could avoid the calcination treatment and reduction treatment necessary in the conventional impregnation method and make Fe catalyst *in situ* originate from metallic Fe (defined as "*in situ* deposition method") rather than Fe oxides on SiO_2. This novel designed process simplifies Fe phase evolution, which is favorable to unravel the influence of the interaction between Fe oxides and SiO_2 on catalyst reduction and subsequent carburization. The different interactions between Fe species (Fe oxides and metallic Fe) and SiO_2 obtained *via* the conventional impregnation method and the *in situ* deposition method will lead to the significant difference in Fe phase composition and Fe nanoparticle morphology during the FTS reaction. For example, the strong Fe oxide-support interaction leads to the difficult reduction and carburization but better resistance to sintering and, however, the weak metallic Fe-support interaction leads to the easy carburization but sintering. This significant difference could provide crucial information in correlating the catalyst structures with the catalytic performance. Besides, based on the experimental results of CO conversion and CO_2 selectivity with time on stream (TOS), the Fe phase composition (Fe_3O_4 and χ-Fe_5C_2), the effects of K promoter and reaction temperature, and the collected amount of H_2O from hot- and ice-cooled traps, the CO_2 formation mechanism in Fe-based FTS reaction is systematically discussed.

2. Experimental

2.1. Catalyst preparation

In order to obtain the *in situ* deposited Fe catalysts, the Fe(CO)*x*-containing syngas (CO/H_2) was firstly prepared by Fe interacting with syngas. In detail, some amount of the fine Fe powder was added into a steel cylinder which was purged with syngas several times. Then, syngas was introduced to pressurize to 10 MPa and kept at the temperature of 60–80 °C for about one month. Prior to reaction, only SiO_2 support was loaded in a fixed-bed reactor. Then, Fe(CO)*x*-containing syngas were introduced into the reactor together under reaction conditions. The Fe(CO)*x* can continuously decompose on SiO_2 to form the supported Fe/SiO_2 catalyst. Particle size of SiO_2 was in the range of 75–125, 125–175 and 420–840 μm, respectively. The SiO_2 particles at 125–175 and 420–840 μm were prepared by firstly grinding the SiO_2 particles (75–125 μm) into fine powder and pressing into a thin cake-like which was then crushed and sieved to the desired size range. The 10Mn-SiO_2 and MnO*x* with 125–175 μm was also used to support the Fe decomposed from Fe(CO)*x* in syngas. The 10Mn-SiO_2 was prepared by the conventional impregnation method. An aqueous solution containing the desired amount of $Mn(NO_3)_2{\cdot}4H_2O$ was added to SiO_2 support and then dried slowly in a rotary evaporator under vacuum at 80 °C for 2 h followed by drying at 120 °C in an oven overnight and calcining in air at 400 °C for 3 h. The manganese oxide (MnO*x*) support was obtained by precipitation of $Mn(NO_3)_2{\cdot}4H_2O$ with $(NH_4)_2CO_3$ as a precipitant. The detailed preparation procedures were same as that in our previous publication [23]. In the case of *in situ* deposited Fe catalysts, it was necessary for us to measure the loading amount of Fe phase on SiO_2 after 60 h reaction. The detailed procedures for measuring Fe content are described as follows: firstly, the as-prepared amount of SiO_2 was loaded into the reactor and then reduced at 350 °C and 0.2 MPa in a flow of H_2 (30 mL min^{-1}) for 3 h. After that, the pretreated SiO_2 was taken out from reactor and weighted as m1. Following the same procedures as above, the Fe(CO)*x*-containing syngas was then fed into the SiO_2-loaded reactor for 60 h FTS reaction under the conditions of 320 °C and 1 MPa. After finishing the reaction, the spent catalyst was taken out and weighted as m2. Thus, the increased weight (m2-m1) was regarded as the Fe phase content. According to the ^{57}Fe Mössbauer spectra showing the Fe phase composition, the SiO_2-supported metallic Fe content could be calculated.

Based on the above-obtained Fe content, 5.3Fe/SiO_2 catalyst having the nominally same metallic Fe content with the sample of deposited spent SiO_2-supported Fe catalyst was prepared by impregnating SiO_2 with $Fe(NO_3)_3{\cdot}9H_2O$ aqueous solution. The drying and calcination conditions were same as the preparation of 10Mn-SiO_2 support. The prepared Fe catalyst and support were labeled as *x*Fe/SiO_2 and *x*Mn-SiO_2, respectively, where *x* indicates the nominal weight loading of Fe (Mn) in the catalyst (support). The calcined 5.3Fe/SiO_2 catalyst was crushed and sieved to 125–175 μm for catalytic tests.100Fe and 99.2Fe0.8 K catalysts were prepared by precipitating iron nitrate with aqueous ammonia solution at pH 8 [24]. The catalysts were labeled as *x*Fe*y*K, where *x* and *y* indicate the nominal weight loading of Fe or K in the catalyst, respectively. The calcined catalyst was crushed and sieved to 40–60 mesh size (about 250–420 μm) for catalytic tests.

2.2. Catalyst characterization

The support SiO_2 and various Fe catalysts were characterized by BET, XRD, HAADF-STEM, ^{57}Fe Mössbauer spectra and TG. The measurement methods are described in detail as follows. Prior to the characterization of spent catalysts, the samples were passivated in a flowing gas of 1%O_2/N_2 at RT for 2 h.

The specific surface area and total pore volume were determined from N_2 adsorption/desorption isotherms at −196 °C using an automated surface area and pore size analyzer (Quantachrome Autosorb iQ). Prior to measurement, the sample was degassed at 150 °C for 4 h under vacuum. The specific surface area was determined according to BET method in the relative pressure range of 0.01-0.3. The pore size and pore volume were calculated based on Barrett-Joyner-Halenda (BJH) method using the desorption branches of N_2 isotherms.

The powder X-ray diffraction (XRD) was recorded by a Bruker AXS D8 Advance diffractometer using Cu(Kα) radiation (λ = 1.5406 Å) at 40 kV and 40 mA at RT with a scanning rate of 1.5° min^{-1} from 5 to 90°. The crystallite phase in sample was determined by comparing XRD patterns with that in standard powder XRD file from International Center for Diffraction Data (JCPDS card).

The HAADF-STEM (high-angle annular dark field scanning transmission electron microscope) observation of spent samples was conducted on a FEI-Tecnai G2 microscope. Before the measurement, the sample was dispersed ultrasonically in ethanol and then deposited on a TEM copper grid.

^{57}Fe Mössbauer spectra were collected in the constant acceleration transmission mode on a Wissel 1550 electromechanical spectrometer (Wissenschaftliche Elektronik GmbH) using a ^{57}Co in Pd matrix irradiation source. The velocity was calibrated by a 25 μm thick α-Fe foil. The isomer shift (IS) was referenced to α-Fe at RT. The spectrum was fitted using a least-squares fitting routine that modeled the spectrum as a combination of singlets, quadruple doublets and magnetic sextets with Lorentzian line shape and constraints in peak width and intensity using MossWinn 3.0i program. The phase composition was derived from the area of absorption peak with assumption of the same recoil-free factor (the probability of absorption of γ photons) for all kinds of iron nuclei in catalyst.

Thermogravimetric (TG) analysis experiments were performed on a Mettler-Toledo TGA-1100SF Thermogravimetric analyzer by heating a sample of about 10 mg from RT to 900 °C at a heating rate of 10 °C min^{-1} in a flow of pure oxygen (50 mL min^{-1}).

2.3. Catalytic performance tests

FTS reactions were carried out in a tubular flow fixed-bed reactor. In general, 3 g of pure support or supported Fe catalyst without dilution by quartz sand was packed into a stainless steel reactor (i.d. of 8.0 mm). It should be especially noted that, in order to check the possibility of Fe (CO)*x* decomposed on the wall of reactor, the Fe(CO)*x*-containing syngas was fed into the tubular reactor without SiO_2 loading to detect the CO conversion under the same operating procedures and reaction conditions. In the case of the precipitated Fe catalysts, such as 100Fe and 99.2Fe0.8 K, about 0.24 g of catalyst was diluted with 13.4 g of quartz sand in order to keep the same catalyst bed height with previously mentioned cases. Before reaction, the sample was reduced at 350 °C and 0.2 MPa in a flow of H_2 (30 mL min^{-1}) for 3 h. Then, the temperature was lowered to the designated reaction temperature (*e.g.*, 280 or 320 °C) under H_2 flow at the same pressure. After increasing the pressure to 1 MPa with syngas ($H_2/CO/N_2$ = 48.5/48.5/3 in volume ratio), the syngas was fed into the reactor at a flow rate of 30 mL min^{-1} to start the FTS reaction.

Gaseous products were hourly analyzed online by a gas chromatography (Agilent GC 7820 A). CO_2, H_2, CO, N_2, CH_4 were measured using a TCD detector with a Porapak Q column and a 5 A molecular sieve column. C_1-C_6 hydrocarbons were analyzed with a Rt-Q-BOND column using an FID detector. 3% of N_2 was used as an internal standard to calculate the CO conversion and product selectivity. The liquid hydrocarbon products collected together from hot and ice-cooled traps were analyzed offline by a gas chromatography GC-2014 AFSC (Shimadzu) equipped with an HP-1 capillary column. The hydrocarbon products and H_2O from hot and ice-cooled traps formed during the FTS reaction were weighed accurately. The CO conversion and product selectivity was reported on a carbon basis expressed in C mol% as presented in our previous publication [41]. Experimental carbon balance in this work is calculated to be at the range of 96–102%.

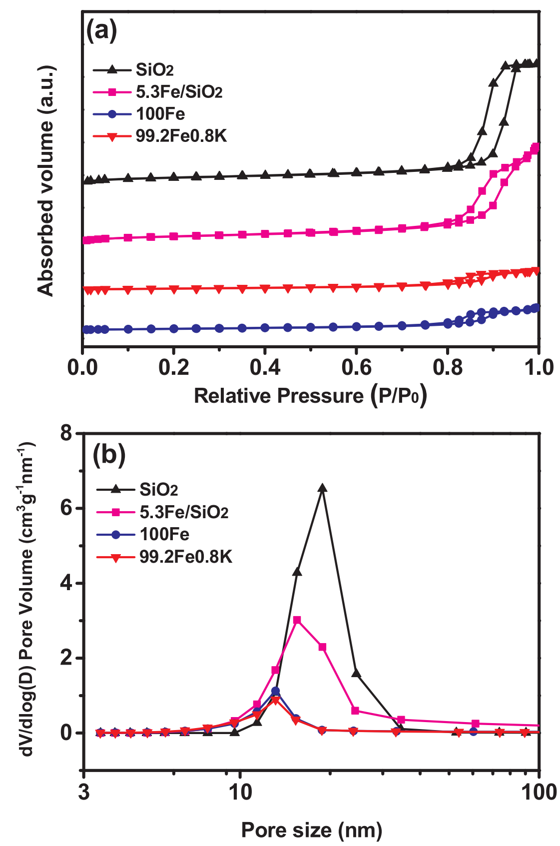


Fig. 1. N_2 adsorption-desorption isotherms (a) and the BJH desorption pore size distribution plots (b) of the fresh samples.

3. Results and discussion

3.1. Characterization of as-prepared SiO_2 support and Fe catalysts prior to the reactions

The textural properties of SiO_2 support and three fresh Fe catalysts are determined by N_2 adsorption/desorption isotherms at −196 °C. The results are summarized in Fig. 1 and Table S1. The tested four samples exhibit an IV-type isotherm according to Brunauer et al [42]. The SiO_2 with average particle size of 75–125 μm has a relatively high surface area of 185.6 m^2/g, large pore volume of 1.16 cm^3/g, and large average pore size of 18.9 nm. The pore size of SiO_2 shows relatively wide mesopores at the range of 10–30 nm. The SiO_2 is used for *in situ* deposition method that Fe(CO)*x* in Fe(CO)*x*-containing syngas is decomposed into metallic Fe on SiO_2 during the FTS reaction. For supported 5.3Fe/SiO_2, the surface area does not alter obviously (189.1 m^2/g) but the pore volume and pore size are slightly lower at 0.93 cm^3/g and 15.5 nm, respectively. Comparing the pore size distribution curves between SiO_2 and 5.3Fe/SiO_2, it indicates that the impregnated Fe species tend to fill into larger pores (Fig. 1b). However, 100Fe prepared *via* precipitation method shows much lower surface area (51.3 m^2/g), lower pore volume (0.22 cm^3/g), and smaller pore size (13.1 nm). The textural properties of 99.2Fe0.8 K are similar to those of 100Fe.

Fig. 2 shows the XRD patterns for the calcined and reduced 5.3Fe/SiO_2 catalysts. The calcined 5.3Fe/SiO_2 displays the weak diffraction

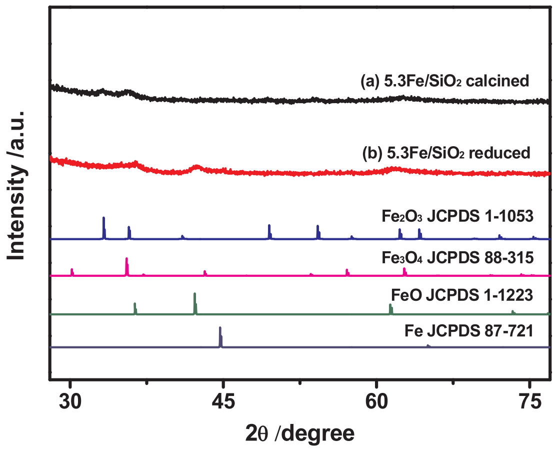


Fig. 2. XRD patterns for the calcined and reduced 5.3Fe/SiO_2 catalysts.

peaks at 33.3° and 35.6° assigned to Fe_2O_3 (JCPDS 1–1053). In contrast, the reduced 5.3Fe/SiO_2 exhibits the diffraction peaks at 36.2°, 42.3°, and 61.3° indicating the presence of FeO (JCPDS 1–1223) rather than Fe_3O_4 and metallic Fe, which suggests that the reduction of FeO into metallic Fe is relatively more difficult.

As shown in Fig. 3a, the Mössbauer spectra could be fitted by one doublet, which indicates only the presence of Fe(III) species in calcined 5.3Fe/SiO_2 (Table S2). The Fe(III) species might exist as the form of iron silicate ($Fe_2(SiO_3)_3$) or ferric oxide (Fe_2O_3). Fig. 3b shows that the blue line is typical of Fe(II) species [43] and the red doublet presents Fe (III) species in reduced 5.3Fe/SiO_2. After the reduction treatment, Fe (III) species are mainly reduced to Fe(II) species which have the content of 60.1% (Table S2). And the part of Fe(III) species could not be reduced due to its existence in the form of iron silicate.

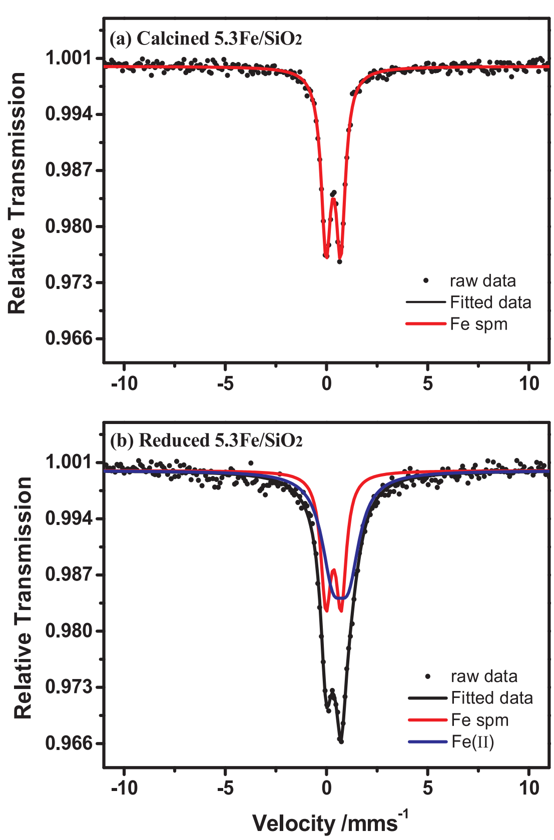


Fig. 3. Mössbauer spectra of the calcined and reduced 5.3Fe/SiO_2 catalysts.

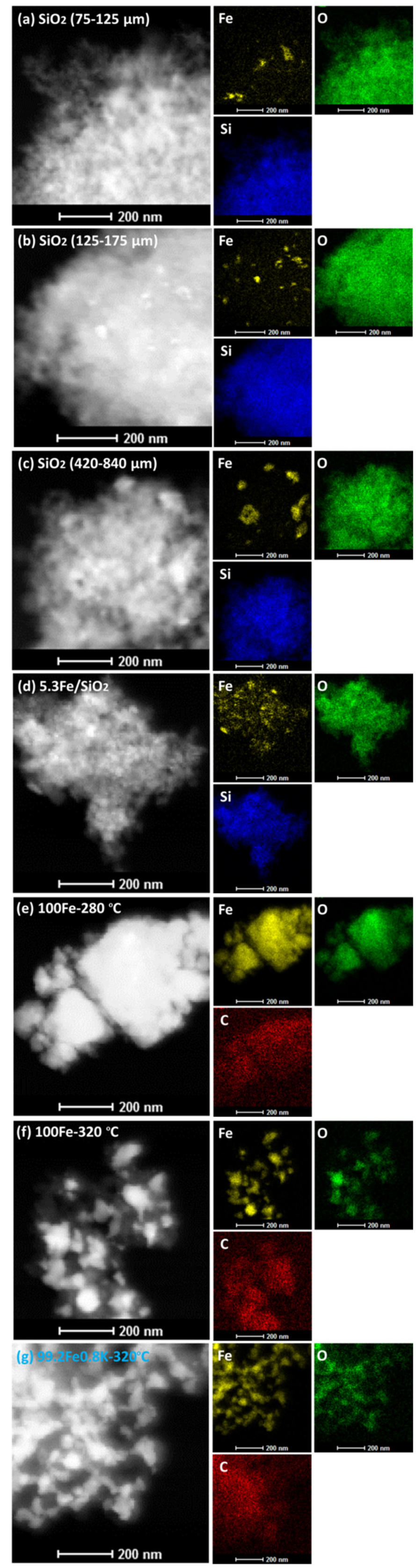


Fig. 4. HAADF-STEM images and the corresponding elemental mapping for the spent Fe catalysts after reaction of 60 h as the corresponding catalytic results are listed in Tables 1,3 and 4. Iron, oxygen, carbon and silicon elements are colored in yellow, green, red and blue, respectively (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

3.2. Characterization of spent Fe catalysts

To get insights into the structural information of spent Fe catalysts prepared with different preparation methods, HAADF-STEM images and the corresponding elemental mapping were employed to characterize these Fe catalysts after 60 h reaction. In the case of *in situ* deposition method, different particle sizes of SiO_2 were applied to study its effect on Fe species distribution during the FTS reaction. It should be noted that SiO_2 with 125–175 μm and 420–840 μm was obtained by crushing SiO_2 particles (75–125 μm) and sieving to the desired size. As shown in Fig. 4a-c, Fe species in yellow color is clearly observed. The Fe species are deposited on SiO_2 to form two different kinds of amorphous. One is that fine Fe NPs are uniformly dispersed on the whole SiO_2 surface. The other is that significantly larger Fe NPs are separately located on SiO_2. Moreover, the larger particle size of SiO_2 facilitates the formation of larger Fe NPs. For the spent 5.3Fe/SiO_2 prepared with the conventional impregnation method (Fig. 4d), Fe species show better dispersion on SiO_2. It should be noted that the nominal Fe content in 5.3Fe/SiO_2 is same with the spent catalyst by *in situ* deposition method with SiO_2 (125–175 μm). In addition, the spent precipitated Fe catalysts (Fig. 4e-g) show obvious carbon accumulation on bulk Fe phase. Upon operating the FTS reaction at lower temperature (280 °C), the carbon deposition is clearly milder as the lighter red color represents the presence of C element in 100Fe-280 °C. The addition of K results in more serious carbon deposition as reflection with deeper red color (99.2Fe0.8 K-320 °C).

Fig. 5 shows the Mössbauer spectra of the abovementioned spent Fe catalysts and the corresponding Mössbauer parameters are compiled in Table S3. The two sextets with Hhf values of 490 and 459 kOe are assigned to the tetrahedral and octahedral sites in Fe_3O_4 [44,45], while the remaining three sextets with Hhf values of 216, 185, and 112 kOe are assigned to three crystallographic sites in χ-Fe_5C_2 [46–49]. For

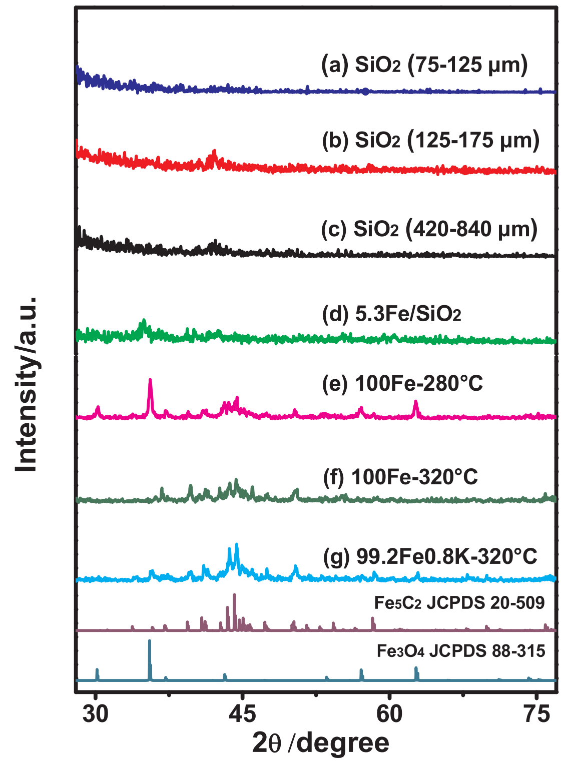


Fig. 6. XRD patterns for the spent Fe catalysts after reaction of 60 h as the corresponding catalytic results are listed in Tables 1, 3 and 4.

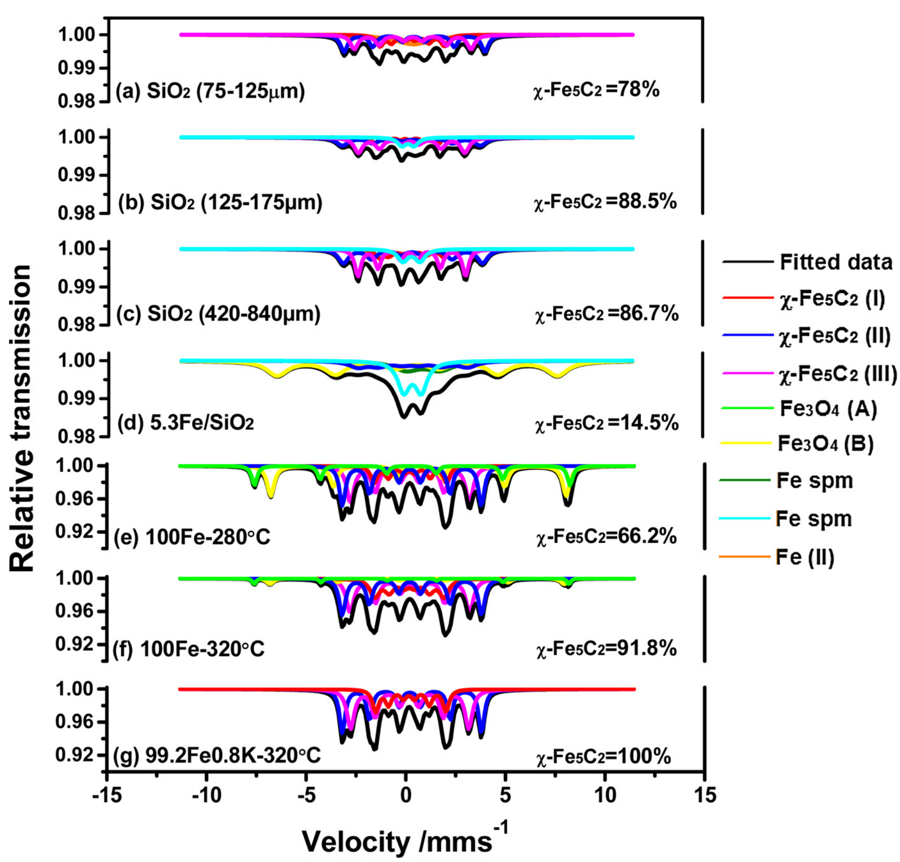


Fig. 5. Mössbauer spectra of the spent Fe catalysts after reaction of 60 h as the catalysts were prepared with different methods and evaluated at different reaction conditions (shown in Tables 1, 3 and 4).

spent SiO_2-supported Fe catalysts (Fig. 5a-d), there is the superparamagnetic Fe(II) and/or Fe(III) species due to the presence of some poorly crystallized iron oxides [50]. Importantly, the content of χ-Fe_5C_2 in spent Fe catalysts is significantly different, which is mainly determined by catalyst preparation method and FTS reaction conditions. The *in situ* deposition method exhibits high χ-Fe_5C_2 content and maximum content (88.5%) is obtained over the SiO_2 (125–175 μm). Surprisingly, the spent 5.3Fe/SiO_2 prepared with the conventional impregnation method results in very low χ-Fe_5C_2 content (14.5%). And the Fe species mainly exists as Fe_3O_4 and superparamagnetic Fe(II) and Fe(III) with the content of 47.0%, 13.9%, and 24.5% (Table S3), respectively. In the case of precipitated 100Fe, a higher reaction temperature facilitates the formation of χ-Fe_5C_2 as the content of χ-Fe_5C_2 increases from 66.2% at 280 °C to 91.8% at 320 °C (Fig. 5e and f). Moreover, the addition of K further promotes the formation of χ-Fe_5C_2 resulting in single χ-Fe_5C_2 phase (Fig. 5g). It should be especially noted that superparamagnetic Fe species does not exist in all spent precipitated Fe catalysts.

Fig. 6 shows the XRD patterns of the spent Fe catalysts. For the *in situ* deposited Fe catalysts, the SiO_2 with particle size of 125–175 μm displays the strongest diffraction peak at 2θ value of 43.2° assignable to χ-Fe_5C_2 (Fig. 6b). In contrast, it seems that no diffraction peak belonging to χ-Fe_5C_2 can be observed for the SiO_2 (75–125 μm). Besides, no diffraction peak at 2θ value of 35.5° assigned to Fe_3O_4 (JCPDS 88–315) were identified for these three spent deposited Fe catalysts (Fig. 6a-c), which is in accordance with Mössbauer spectra (Fig. 5a-c). Compared to the spent deposited Fe catalysts, the impregnated 5.3Fe/SiO_2 exhibits intensive diffraction peak at 2θ value of 35.3° assignable to Fe_3O_4 and significantly weaker peaks for χ-Fe_5C_2 (Fig. 6d), which is in well agreement with Mössbauer parameters as the content of Fe_3O_4 and χ-Fe_5C_2 is 47% and 14.5% (Table S3), respectively. Over three spent precipitated Fe catalysts, the 100Fe-280 °C shows the strongest diffraction peak for Fe_3O_4 at 2θ value of 35.5° and relatively weak diffraction peaks at 2θ values of 43.4° and 44.1° assigned to the χ-Fe_5C_2 (JCPDS 20–509) (Fig. 6e). The peaks at 43.4° and 44.1° are attributed to (021) and (510) facet of χ-Fe_5C_2, respectively. The diffraction peaks for χ-Fe_5C_2 are clearly strengthened for 100Fe-320 °C (Fig. 6f) and further more prominent with the addition of K (99.2Fe0.8 K-320 °C) (Fig. 6g). Correspondingly, the diffraction peaks for Fe_3O_4 do gradually disappear with an increase in the content of χ-Fe_5C_2.

Fig. 7 shows the TG profiles of the spent Fe catalysts. 100Fe-280 °C displays the continuously increased weight in the whole temperature range from 50 to 900 °C (Fig. 7a). The 100Fe-280 °C contains Fe_3O_4 (33.8%) and χ-Fe_5C_2 (66.2%) (Table S3). As Fe_3O_4 and χ-Fe_5C_2 are converted into α-Fe_2O_3 during the TG measurement with O_2, the weight could increase by 3.5% and 31.6%, respectively. The weight increased is due to less amount of carbon deposition during the FTS reaction at low temperature (280 °C). For spent 5.3Fe/SiO_2, no obvious weight loss is observable over the whole TG curve (Fig. 7b), which is ascribed to the low Fe content and the high proportion of Fe_3O_4 rather than χ-Fe_5C_2. Different from the above cases, 100Fe-320 °C shows a rapid weight increase in the temperature range of 50–298 °C, which results from the oxidation of χ-Fe_5C_2 into α-Fe_2O_3. When the temperature elevates to 298 °C, the remarkable weight loss starts and terminates in a very narrow temperature range (Fig. 7c). In contrast, 99.2Fe0.8 K-320 °C shows a similar weight loss curve and, however, the weight loss starts at a lower temperature of 246 °C (Fig. 7d). The weight loss at a lower temperature indicates that the oxidation of deposited carbon species proceeds more easily over the K-promoted Fe catalyst. Compared the TG profiles of four samples, it is clear that the high reaction temperature and the K promoter could effectively facilitate CO dissociation leading to more serious carbon accumulation.

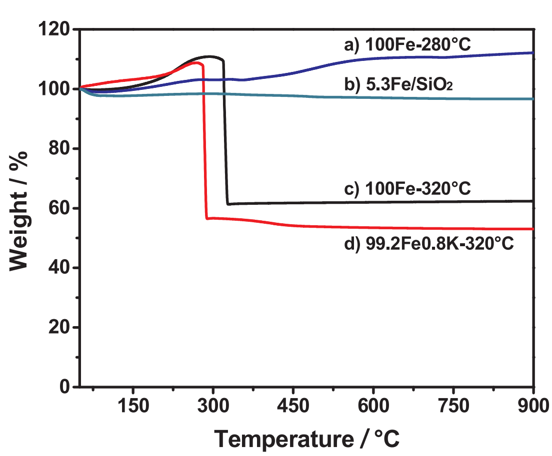


Fig. 7. TG profiles for the spent Fe catalysts after reaction of 60 h as the catalytic results are shown in Tables 3 and 4.

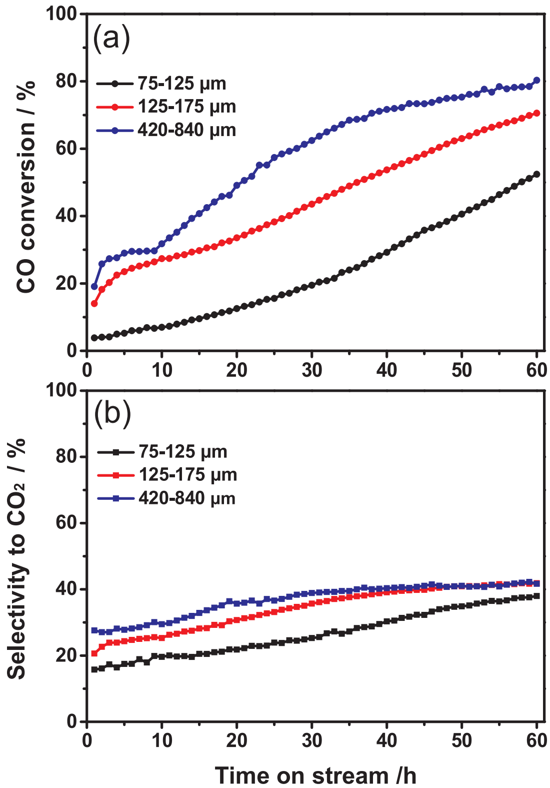


Fig. 8. Effect of the SiO_2 particle size on the CO conversion and the selectivity to CO_2 with TOS over *in situ* deposited Fe catalysts.

3.3. Catalytic results

Fig. 8 presents the effect of SiO_2 particle size on the CO conversion and the selectivity to CO_2 with TOS over *in situ* deposited Fe catalysts. One can see that CO conversion is continuously increased during the whole FTS reaction. And the larger SiO_2 particle size, the higher CO conversion. Concerning the selectivity to CO_2, it also shows the tendency of increase with increased CO conversion. When correlating CO conversion with CO_2 selectivity, it seems that CO_2 selectivity increases more rapidly as CO conversion is lower than about 50%. The average CO conversions and product selectivities in 60 h reaction are summarized in Table 1. Since the Fe(CO)x is continuously introduced into the reactor with syngas feed, it is difficult to obtain the stable CO conversion and product selectivities. It is reasonable for us to evaluate the reaction performance based on the average values of the whole reaction period. As mentioned above, the larger SiO_2 particle size corresponds to the higher average CO conversion and selectivity to CO_2. Despite that

Table 1
Effect of the SiO_2 particle size on the catalytic performance over *in situ* deposited Fe catalysts.[a]

Particle size of SiO_2 (μm)	CO conversion (%)	Selectivity to CO_2 (%)	Theoretically minimum amount H_2O_{min}/g [b]	Theoretically maximum amount H_2O_{max}/g [c]	Experimentally collected amount H_2O_{exp}/g [d]	Selectivity to hydrocarbons (%)				O/(O + P)[e] (%)
						CH_4	$C_2^{=}$-$C_4^{=}$	C_2°-C_4°	C_{5+}	
75-125	23.2	26.6	0	9.77	4.07	15.9	22.8	7.4	53.9	75.6
125-175	44.6	34.2	0	18.77	3.74	15.6	18.2	9.8	56.3	65.4
420-840	57.8	36.7	0	24.32	3.13	16.8	16.3	12.2	54.7	57.7

[a] Reaction conditions: 320 °C, 1.0 MPa, loading of SiO_2 = 3 g, H_2/CO = 1, syngas flow rate = 30 mL/min, reaction time = 60 h, the results are the average values in the reaction period of 60 h as calculated according to the equation: $\bar{X} = \frac{\sum_i^{60} X_i}{60}$; X_i is the value of CO conversion or product selectivity.

[b] theoretically minimum amount of H_2O by assuming all O* species react with CO form CO_2 *via* the Boudouard reaction.

[c] theoretically maximum amount of H_2O by assuming all O* species undergo hydrogenation to form H_2O.

[d] experimentally collected amount of H_2O during the FTS reaction of 60 h.

[e] the percentage of olefin to (olefin + paraffin) in C_2-C_4 hydrocarbons.

the selectivities to CH_4 and C_{5+} do not vary substantially, the olefin-to-(olefin + paraffin) ratio (O/(O + P)) in C_2-C_4 hydrocarbons is significantly changed as the lower CO conversion gives the higher selectivity to lower olefins ($C_2^{=}$-$C_4^{=}$). In this study, we defined two theoretical values for studying the formation of CO_2 and H_2O. The minimum value H_2O_{min} in Table 1 is the theoretical amount of collected H_2O by assuming all O* species from CO dissociation react with CO to form CO_2 *via* the Boudourd reaction. With this assumption, the amount of collected H_2O_{min} is 0 g. The maximum value H_2O_{max} in Table 1 is the theoretical amount of collected H_2O by assuming all O* species undergo hydrogenation to form H_2O. Based on this assumption, the amount of collected H_2O_{max} could be calculated from the amount of O* species through CO conversion. The experimentally collected amount H_2O_{exp} is defined as H_2O obtained from hot- and ice-cooled traps after the FTS reaction. Seen from data in Table 1, it is clear that there is a trend that CO_2 selectivity increases with decreased H_2O_{exp}. More definitely, the higher selectivity to CO_2 results in the lower H_2O_{exp} which is closer to H_2O_{min} and deviates far away from H_2O_{max}.

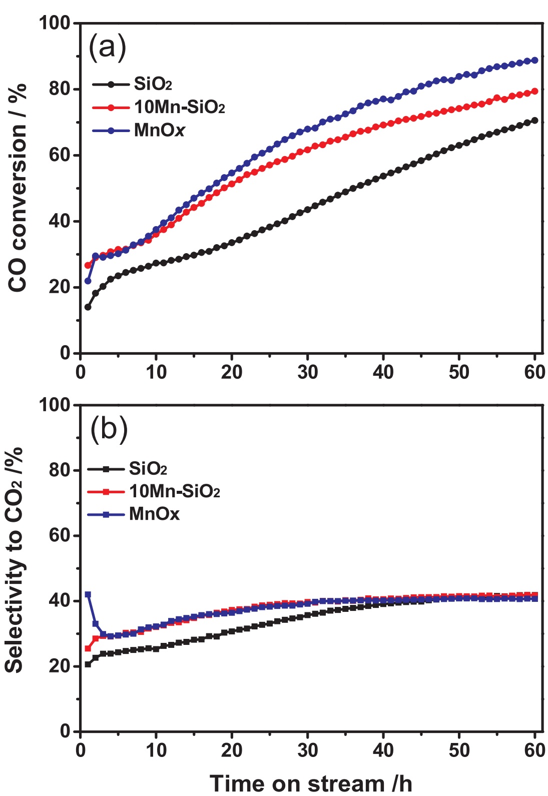


Fig. 9. Effect of the support character on the CO conversion and the selectivity to CO_2 with TOS over *in situ* deposited Fe catalysts.

Fig. 9 shows the effect of support character on the CO conversion and the selectivity to CO_2 with TOS over *in situ* deposited Fe catalysts. In contrast to un-modified SiO_2, the manganese-modified SiO_2 (10Mn-SiO_2) and MnO*x* support exhibit a similar trend of continuously increased CO conversion during the FTS reaction. However, the presence of Mn obviously promotes CO conversion with a milder increase in CO_2 selectivity with TOS. As shown in Table 2, MnO*x* indicates the highest average CO conversion (63.6%) and the highest average CO_2 selectivity (38.3%). Different from the effect of SiO_2 particle size, the presence of Mn does not obviously alter the hydrocarbon distribution and O/(O + P). Similar to previous results in Table 1, the higher average CO conversion corresponds to the higher selectivity to CO_2, which also results in the lower H_2O_{exp}.

Fig. 10 shows the effect of catalyst preparation method on the CO conversion and the selectivity to CO_2 with TOS over as-prepared three catalysts. Different of the effect of SiO_2 particle size and support character, the catalyst preparation method gives rise to the significant difference in CO conversion with TOS during the FTS reaction (Fig. 10a). The supported 5.3Fe/SiO_2 prepared with the conventional impregnation method also indicates the continuously increased CO conversion with a milder trend and a much lower CO conversion (49.8%) at the end of 60 h in contrast to the *in situ* deposited Fe catalyst with SiO_2 (125–175 μm) having a remarkably higher CO conversion (70.5%). However, the 100Fe prepared with the precipitated method results in a totally different trend in CO conversion. The CO conversion is initially rapidly increased to reach maximum value and then drops sharply to the lowest value followed by gradually elevated CO conversion with TOS till the end of FTS reaction. As shown in Fig. 10b, in spite of the significantly different trend in CO conversion with TOS, the selectivity to CO_2 is basically closely related with CO conversion as reflected by previously mentioned cases. Furthermore, seen from Fig. 10a and b, it is easy to observe that the *in situ* deposited Fe catalyst shows the higher selectivity to CO_2 compared to the impregnated 5.3Fe/SiO_2 as the same CO conversion at 38.3% corresponds to CO_2 selectivity at 33.1% and 27.6%, respectively. This result might suggest that CO_2 selectivity is related to not only CO conversion but also Fe phase composition such as the content of χ-Fe_5C_2 and Fe_3O_4. Table 3 shows the average catalytic results in 60 h. It is clear that the average CO conversion over the *in situ* deposited Fe catalyst is slightly higher than 5.3Fe/SiO_2. However, it should be noted that the Fe content gradually increases with TOS over the *in situ* deposited Fe catalyst. And the nominal Fe content after 60 h is same as that of 5.3Fe/SiO_2. Therefore, it is more reasonable to compare CO conversion at the 60th hour. From this viewpoint, the *in situ* deposited Fe catalyst shows much higher activity than the impregnated 5.3Fe/SiO_2. In addition, both the O/(O + P) in C_2-C_4 hydrocarbons and C_{5+} selectivity are clearly higher than that for 5.3Fe/SiO_2 and 100Fe.

Fig. 11 shows the effect of reaction temperature and K promoter on the CO conversion and the selectivity to CO_2 with TOS over the precipitated Fe catalysts. The results for 100Fe-320 °C could be used as reference to investigate the effect of reaction temperature and K

Table 2
Effect of the support character on the catalytic performance over *in situ* deposited Fe catalysts.[a]

Support[b]	CO conversion (%)	Selectivity to CO_2 (%)	Theoretically minimum amount H_2O_{min}/g [c]	Theoretically maximum amount H_2O_{max}/g [d]	Experimentally collected amount H_2O_{exp}/g [e]	Selectivity to hydrocarbons (%)				O/(O + P)[f] (%)
						CH_4	$C_2^=$-$C_4^=$	C_2°-C_4°	C_{5+}	
SiO_2	44.6	34.2	0	18.77	3.74	15.6	18.2	9.8	56.3	65.4
10Mn-SiO_2	58.1	37.9	0	24.45	3.20	17.2	16.4	12.4	54.0	57.2
MnOx	63.6	38.3	0	26.77	2.64	13.6	17.6	10.0	58.8	63.7

[a] Reaction conditions: 320 °C, 1.0 MPa, loading of SiO_2 = 3 g, H_2/CO = 1.0, syngas flow rate = 30 mL/min, reaction time = 60 h, the results are the average values in the reaction period of 60 h as calculated according to the equation shown in Table 1.
[b] The particle size is 125–175 μm; [c–f] The same as that in Table 1b–e.

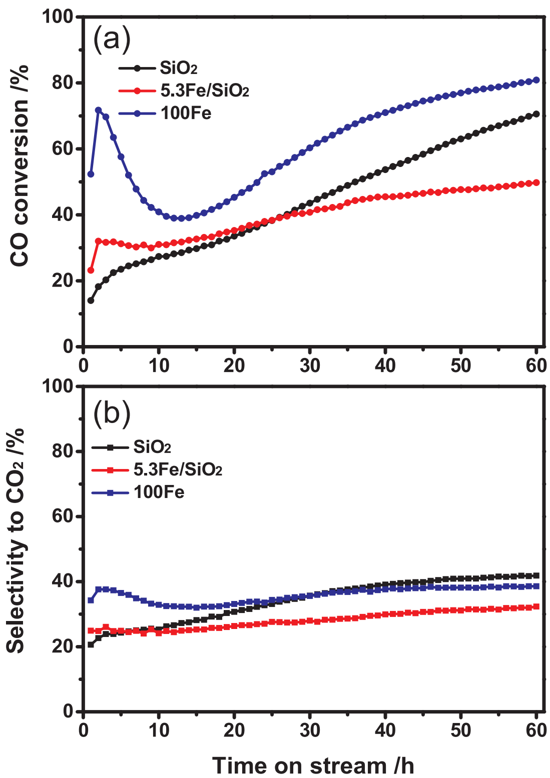


Fig. 10. Effect of the catalyst preparation method on the CO conversion and the selectivity to CO_2 with TOS over as-prepared catalysts.

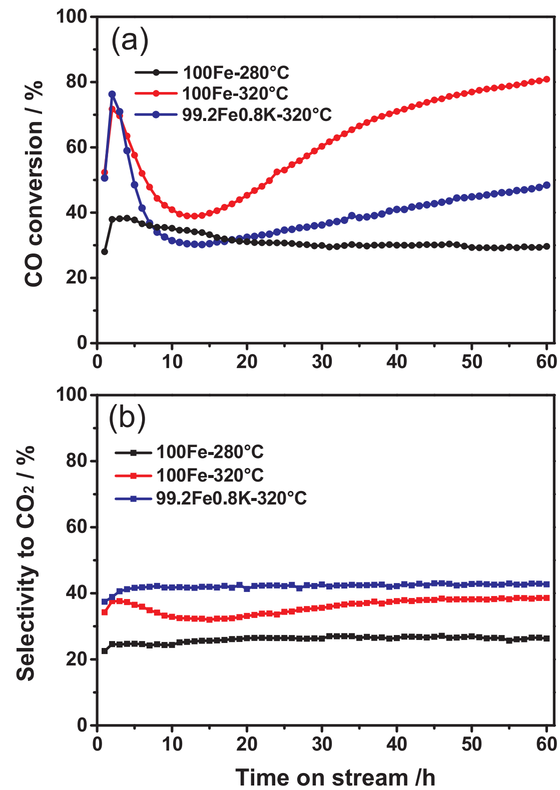


Fig. 11. Effect of the reaction temperature and the K promoter on the CO conversion and the selectivity to CO_2 with TOS over the precipitated Fe catalysts.

promoter. Seen from Fig. 11a, when operating the FTS reaction at 280 °C, CO conversion is continuously decreased with a slow deactivation rate in the whole reaction of 60 h and does not show a rapid decline in the early reaction stage. As CO conversion becomes relatively stable over 100Fe-280 °C, CO_2 selectivity is also kept stable and notably lower than 100Fe-320 °C. For the K-promoted 99.2Fe0.8 K-320 °C, CO conversion exhibits a similar trend to 100Fe-320 °C. However, CO conversion drops more dramatically at initial reaction stage and then increases with a milder manner at subsequent reaction period, which is mainly due to the promoted carbon deposition by K. Interestingly, although CO conversion varies dramatically with TOS, CO_2 selectivity is quite stable over 99.2Fe0.8 K-320 °C. Besides, different from results in Table 1, 99.2Fe0.8 K-320 °C shows obviously lower CO conversion in

Table 3
Effect of the catalyst preparation method on the catalytic performance over as-prepared catalysts.[a].

Catalyst	CO conversion (%)	Selectivity to CO_2 (%)	Theoretically minimum amount H_2O_{min}/g [e]	Theoretically maximum amount H_2O_{max}/g [f]	Experimentally collected amount H_2O_{exp}/g [g]	Selectivity to hydrocarbons (%)				O/(O + P)[h] (%)
						CH_4	$C_2^=$-$C_4^=$	C_2°-C_4°	C_{5+}	
SiO_2[b]	44.6	34.2	0	18.77	3.74	15.6	18.2	9.8	56.3	65.4
5.3Fe/SiO_2[c]	40.2	28.1	0	16.92	4.24	17.7	18.3	16.2	47.8	53.2
100Fe[d]	61.6	35.9	0	25.93	3.66	18.1	19.0	13.7	49.1	58.1

[a] Reaction conditions: 320 °C, 1.0 MPa, H_2/CO = 1.0, syngas flow rate = 30 mL/min, reaction time = 60 h, the results are the average values in the reaction period of 60 h as calculated according to the equation shown in Table 1.
[b] Particle size = 125–175 μm, support loading = 3 g.
[c] Particle size = 125–175 μm, catalyst loading = 3 g.
[d] Particle size = 250–420 μm, catalyst loading = 0.24 g; [e–h] The same as that in Table 1b–e.

Table 4
Effect of the reaction temperature and the K promoter on the catalytic performance over the precipitated Fe catalysts.[a]

Catalyst[b]	Reaction temperature (°C)	CO conversion (%)	Selectivity to CO_2 (%)	Theoretically minimum amount H_2O_{min}/g [c]	Theoretically maximum amount H_2O_{max}/g [d]	Experimentally collected amount H_2O_{exp}/g [e]	Selectivity to hydrocarbons (%)				O/(O + P)[f] (%)
							CH_4	$C_2^=$-$C_4^=$	$C_2^°$-$C_4^°$	C_{5+}	
100Fe	280	31.4	26.0	0	13.21	4.91	13.6	14.3	17.3	54.9	45.3
100Fe	320	61.6	35.9	0	25.93	3.66	18.1	19.0	13.7	49.1	58.1
99.2Fe0.8 K	320	40.4	42.1	0	17.00	0.88	18.5	17.0	6.8	57.7	71.6

[a] Reaction conditions: 1.0 MPa, H_2/CO = 1.0, catalyst loading = 0.24 g, syngas flow rate = 30 mL/min, reaction time = 60 h, the results are the average values in the reaction period of 60 h as calculated according to the equation shown in Table 1.
[b] Particle size = 250–420 μm; [c–f] The same as that in Table 1b–e.

the whole reaction of 60 h compared to 100Fe-320 °C. However, CO_2 selectivity is clearly higher. This result suggests that K promoter enhances the formation of CO_2. Table 4 lists the average catalytic results in 60 h. It is clear that the higher reaction temperature results in increased CO conversion and higher CO_2 selectivity. In the case of 99.2Fe0.8 K-320 °C, CO conversion is remarkably lower relative to 100Fe-320 °C due to the resulting deactivation from carbon deposition. 99.2Fe0.8 K-320 °C shows the highest CO_2 selectivity, which corresponds to the lowest H_2O_{exp} (0.88 g). It seems that H_2O_{exp} mainly depends on CO_2 selectivity rather than CO conversion.

3.4. Discussion

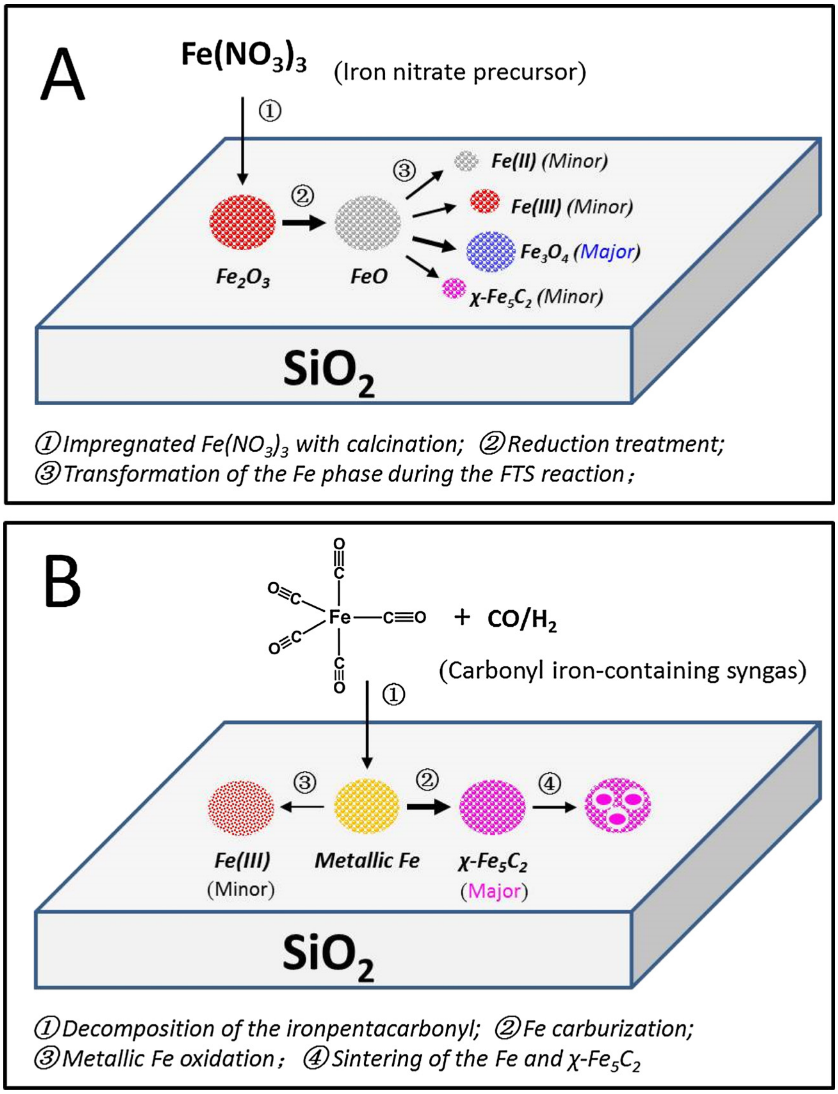
The characterization results and the corresponding catalytic performance over various as-prepared Fe catalysts have been systematically presented. Based on these results, it is informative to collect reliable evidences for unveiling crucial aspects in determining the catalytic performance and bringing in-depth molecular insights into FTS catalysis. Therefore, the discussions will focus on the following concerns.

3.4.1. Significant difference in Fe phase evolution between the conventional impregnation method and the in situ deposition method

Seen from illustration in Scheme 1, it is obvious that the final composition and morphology of Fe phase over spent catalysts are totally different, which is strongly dependent of the catalyst preparation method. Indeed, the preparation method determines the kind of starting iron precursor and its existing chemical environment directly leading to the different evolution process of Fe species.

In the case of impregnated 5.3Fe/SiO_2, $Fe(NO_3)_3$ is selected as iron precursor. The precursor must decompose into Fe_2O_3 regardless of the calcination treatment in air or inert gas (N_2) as shown in Scheme 1A (pathway ①). This result is well confirmed by XRD patterns (Fig. 2a) and Mössbauer spectra (Fig. 3a). The calcined Fe_2O_3 should have strong interaction with SiO_2 over 5.3Fe/SiO_2. The strong interaction between Fe_2O_3 and SiO_2 is not favorable for its reduction into metallic Fe which could be easily carburized into active iron carbide phase in FTS reaction. Indeed, Fe_2O_3 could not be reduced into metallic Fe but only into FeO during the reduction treatment (Scheme 1A (pathway ②)) as demonstrated by XRD patterns (Fig. 2b). It should be noted that for calcined 5.3Fe/SiO_2, the calcination treatment leads to the formation of $Fe_2(SiO_3)_3$ which exists as Fe(III) and is difficult to be reduced during the reduction treatment (Fig. 3b and Table S2) [51–53]. However, with a long reaction time in syngas and a relatively high reaction temperature (320 °C), it could further evolve in part into Fe_3O_4, which is demonstrated by Mössbauer parameters as Fe(III) content obviously decreases from 39.9% to 24.5% (Table S2). Meanwhile, it is also difficult for FeO to carburize into χ-Fe_5C_2 (Scheme 1A (pathway ③)). Therefore, the χ-Fe_5C_2 content in spent 5.3Fe/SiO_2 is as low as 14.5% (Fig. 12). In the FTS reaction, the formation of oxidative H_2O could oxidize FeO into Fe_3O_4. As a result, Fe_3O_4 presents the highest content (47.0%), which is also due to the $Fe_2(SiO_3)_3$ evolution into Fe_3O_4. And Fe(II) and Fe(III) exist with minor contents of 13.9% and 24.5%, respectively. It should be noted that over spent 5.3Fe/SiO_2, the Fe(III) content mainly comes from the difficultly reduced $Fe_2(SiO_3)_3$ and the Fe(II) is ascribed to difficultly carburized FeO. The strong interaction of Fe oxides with SiO_2 leads to difficult reduction to metallic Fe, which is directly responsible for Fe phase composition character over spent 5.3Fe/SiO_2. However, the strong Fe oxides-SiO_2 interaction could hinder the aggregation of Fe nanoparticles during the FTS reaction. As shown in Fig. 4d, the HAADF-STEM image shows that Fe species is well dispersed in spent 5.3Fe/SiO_2, which indicates that the sintering of Fe nanoparticles does not occur easily during the FTS reaction.

In contrast to the conventional impregnation method, the newly designed *in situ* deposition method is totally different in catalyst fabrication and Fe phase evolution. Definitely, over the *in situ* deposition method, the catalyst fabrication and FTS reaction proceed concurrently. Firstly, $Fe(CO)x$-containing syngas is fed into reactor and $Fe(CO)x$ directly decomposes into metallic Fe on SiO_2 with the releasing of CO rather than to form Fe oxides (Scheme 1B (pathway ①)), which has been demonstrated by Ding Ma's group as they prepared metallic Fe NPs by decomposing the $Fe(CO)_5$ in nitrogen at 180 °C [31]. The absence of Fe oxides in the *in situ* deposition process could avoid the influence of strong Fe oxides-SiO_2 interaction which results in poor reducibility. Moreover, the formed metallic Fe on SiO_2 could be easily carburized into χ-Fe_5C_2 in syngas (Scheme 1B (pathway ②)). It should be noted that under the high reaction temperature and the relatively high CO concentration (CO/H_2 = 1) it is possibly difficult for us to detect the presence of transitional metallic Fe phase because of its immediate transformation into iron carbide. As a result, the χ-Fe_5C_2 content in spent catalyst is as high as 88.5% (Fig. 12). At the same time, the metallic Fe (or χ-Fe_5C_2) could also be oxidized into Fe(III) with a minor pathway (Scheme 1B (pathway ③)) due to the oxidative H_2O produced in FTS reaction. Thus, Fe(III) content in spent catalyst is as low as 11.5% (Fig. 12). It should be noted that over the *in situ* deposited Fe catalyst, it is not observable for Fe_3O_4 which is major component in spent 5.3Fe/SiO_2. Different from the impregnated Fe oxide on SiO_2, the *in situ* deposition method directly forms the metallic Fe on SiO_2. The metallic Fe has weak interaction with SiO_2, which leads to easy sintering of small metallic Fe and χ-Fe_5C_2 nanoparticles (Scheme 1B (pathway ④)) as reflected in Fig. 4c. From the elemental mapping of spent *in situ* deposited Fe on SiO_2, it is clear that Fe nanoparticles are divided into two kinds. One is fine Fe nanoparticles which are uniformly distributed on SiO_2. The other is large Fe nanoparticles which are discretely dispersed on SiO_2. These large Fe nanoparticles might be formed by aggregation of initially deposited fine Fe nanoparticles. In this part, we have clearly discussed the evolution process of Fe phase over the different catalyst preparation method by combining the various catalyst characterizations and processing analysis. In addition, it should be mentioned that for iron carbide phase, only χ-Fe_5C_2 was observed over the samples prepared by the different methods, which has been well characterized by the XRD patterns and Mössbauer parameters as shown in Figs. 3, 5 and 6, and Tables S2 and S3.



Scheme 1. The schematic illustration of the evolution process of Fe phase over the impregnated 5.3Fe/SiO_2 catalyst (A) and the *in situ* deposited Fe on SiO_2 (B) during the catalyst preparation, reduction treatment, and FTS reaction.

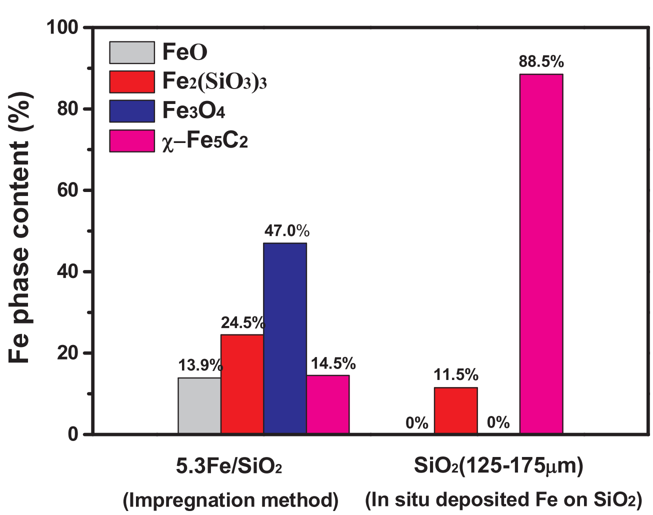


Fig. 12. The comparison of Fe phase composition between the spent 5.3Fe/SiO_2 and the spent *in situ* deposited Fe catalyst.

3.4.2. Relationship between the Fe catalyst structure and the catalytic performance

As shown in Fig. 8a and Table 1, the particle size of SiO_2 displays the remarkable effect on the CO conversion with TOS and the average CO conversion. It is clear that the smaller particle size leads to a notably lower catalytic activity. This might be caused by the catalyst fabrication processing in FTS reactor. Different from the fluidized bed, SiO_2 support is kept statically in the fixed-bed reactor. When the particle size is very small to show narrow interspace among the particles, Fe(CO)*x* precursor is easier to deposit on top of SiO_2 bed leading to less amount of the deposited Fe on SiO_2 and thus the formed Fe phase shows poorer dispersion to give lower CO conversion with the smaller SiO_2 particle size. In other words, the decomposed Fe tends to aggregate since Fe (CO)*x* could not enter into inner SiO_2 bed smoothly. Reasonably, with an increase in the SiO_2 particle size, the Fe(CO)*x* species are easier to enter into the whole SiO_2 bed, which results in more active sites formed on SiO_2 support and higher catalytic activity. It should be noted that for the *in situ* deposition method, the amount of deposited Fe continuously increases with increased TOS. To exclude the effect of possible deposition of Fe(CO)*x* on the wall of reactor, the blank experiment was carried out without SiO_2 loading under the same operating procedures and reaction conditions. The online GC data shows almost no CO conversion, which suggests that the Fe(CO)*x* species have not dominantly

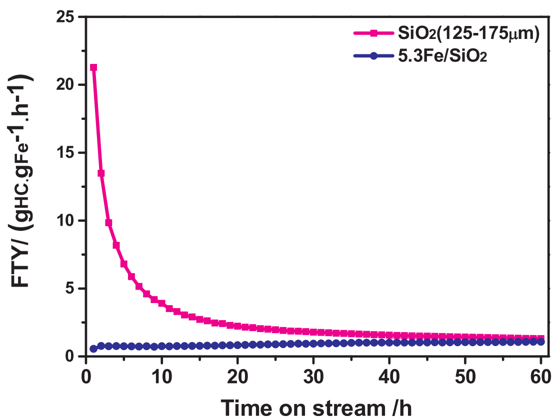


Fig. 13. The FTY (iron time yield) of 5.3Fe/SiO_2 and *in situ* deposited Fe catalyst with TOS.

deposited on the wall of reactor due to its easy flowing through the reactor without obstruction. From this result, it could be deduced that the decomposed Fe as the active sites on the wall of reactor does not show a considerable impact on converting syngas. Since Fe(CO)x is directly deposited on SiO_2 to form metallic Fe, it is easily transformed into active χ-Fe_5C_2. Therefore, the increased CO conversion with TOS might mainly be ascribed to the increased Fe content on SiO_2 rather than the slow carburization process in the formation of χ-Fe_5C_2. Based on the above discussion, it is reasonable that the larger particle size of SiO_2 should have higher catalytic activity due to more amount of deposited Fe on SiO_2.

In the case of impregnated 5.3Fe/SiO_2, CO conversion shows the continuously slow increase in 60 h reaction (Fig. 10a). The increased activity with TOS might be mainly due to slow reduction and subsequent carburization process of FeO in syngas (Fig. 2b and Fig. 5d). As mentioned previously, for 5.3Fe/SiO_2, all Fe species are firstly introduced into SiO_2 support and then need activation in H_2 prior to FTS reaction. For *in situ* deposition method, Fe(CO)x precursor is continuously introduced into a fixed-bed reactor during the FTS reaction. Therefore, the Fe content increases with a longer reaction time. Seen from FTY (iron time yield) shown in Fig. 13, the deposited Fe on SiO_2 shows a super high FTY value (21.3 $g_{HC}{\cdot}g_{Fe}{}^{-1}{\cdot}h^{-1}$) at the first hour. Then, FTY value rapidly declines at the early reaction stage with TOS. It is imaginable that Fe(CO)x is initially decomposed to metallic Fe and quickly carburized to fine χ-Fe_5C_2 nanoparticles uniformly on SiO_2, which results in high FTY value. With FTS reaction progress, the Fe content on SiO_2 continuously increases. In spite of increased Fe content, the weak interaction between χ-Fe_5C_2 (or metallic Fe) and SiO_2 support results in easy sintering (Scheme 1B (pathway ④)), which gives lower FTY value with increased TOS [15,54]. Undoubtedly, the smaller χ-Fe_5C_2 nanoparticles formed at the early reaction stage leads to an easier sintering corresponding to more rapid declining in the FTY value (Fig. 13). At the end of reaction (60 h), FTY value for *in situ* deposited Fe catalyst is about 1.3 $g_{HC}{\cdot}g_{Fe}{}^{-1}{\cdot}h^{-1}$ slightly higher than 1.1 $g_{HC}{\cdot}g_{Fe}{}^{-1}{\cdot}h^{-1}$ for impregnated 5.3Fe/SiO_2 (Fig. 13). Although the active χ-Fe_5C_2 content (88.5%) in spent *in situ* deposited catalyst is much higher than that (14.5%) for spent impregnated 5.3Fe/SiO_2 (Fig. 12), the conventional impregnation method makes better dispersion of χ-Fe_5C_2 due to stronger metal-support interaction. Therefore, FTY values are close in spite of significantly different Fe phase composition between these two spent Fe catalysts.

Fig. 11a shows the effect of reaction temperature and K promoter on the CO conversion with TOS over precipitated Fe catalysts. Interestingly, only the catalytic test at 280 °C (100Fe-280 °C) indicates the gradual decrease in CO conversion in the whole reaction of 60 h. This result might be mainly result from slower CO dissociation at lower temperature which does not favor the carburization to form active χ-Fe_5C_2. Indeed, the spent 100Fe-280 °C shows much lower χ-Fe_5C_2 content (66.2%) compared to spent 100Fe-320 °C (91.8%) (Fig. 5e and f). The low χ-Fe_5C_2 content suggests that the gradual deactivation might not be due to carbon deposition covering active sites but due to oxidation of χ-Fe_5C_2 to Fe_3O_4. This viewpoint could be confirmed by TG profile and elemental mapping over spent 100Fe-280 °C (Figs. 7a and 4 e). It is clear that there has no weight loss in the range of 50 to 900 °C. As shown in Table 4, in spite of lower average CO conversion (31.4%), H_2O_{exp} is the highest (4.91 g) over 100Fe-280 °C, which no doubt favors the oxidation of χ-Fe_5C_2.

In the case of 100Fe-320 °C, the relatively high reaction temperature remarkably promotes CO dissociation at initial reaction stage. The dissociative C* from CO could not timely be converted into FTS products *via* its hydrogenation and thus must cover the active sites to result in rapid drop in CO conversion (Fig. 11a). These dissociative C* species are actually active as confirmed by TG profile (Fig. 7c), which could be easily removed completely at relatively low temperature of about 334 °C. Therefore, these C* species could carburize reduced metallic Fe to increase the χ-Fe_5C_2 content which results in gradual increase in CO conversion with TOS. As expected, the spent 100Fe-320 °C has a high χ-Fe_5C_2 content (91.8%) (Fig. 5f). When K promoter is added into the catalyst, 99.2Fe0.8 K-320 °C exhibits more serious deactivation at initial reaction stage, which results from K-promoted carbon deposition covering active sites. K could also facilitate the carburization of reduced metallic Fe into χ-Fe_5C_2 during the FTS reaction. Indeed, the χ-Fe_5C_2 content in spent 99.2Fe0.8 K-320 °C is as high as 100% (Fig. 5g). However, CO conversion over 99.2Fe0.8 K-320 °C is always lower than that over 100Fe-320 °C, which suggests that more serious carbon deposition is responsible for lower CO conversion as demonstrated by TG profiles in Fig. 7c and d.

3.4.3. Mechanistic insights into the formation pathway of CO_2 and the factor in determining the selectivity of CO_2

Different from Co-based FTS, a large amount of CO_2 is produced along with CO hydrogenated into hydrocarbons, which could substantially reduce the utilization efficiency of C atom. Therefore, great efforts have been taken to reduce CO_2 selectivity over Fe-based FTS. However, the surface reaction to affect CO_2 selectivity is very complicated over Fe catalysts, which leads to the difficulty in obtaining the atomic-scale insights into this important subject. To be skilled in regulating CO_2 selectivity, it should understand the formation pathway of CO_2 and the main factor in determining its selectivity. In our previous investigation, the Boudouard mechanism plays a predominant role in CO_2 formation on active χ-Fe_5C_2, while the hydrogenation of surface O* to form H_2O is unfavorable, which has been well illustrated by combining experiments and DFT calculations [41]. It should be noted that CO_2 formation is also possible *via* the WGS reaction by reacting CO molecules with the H_2O formed on χ-Fe_5C_2. The existence of Fe_3O_4 favors the reverse water-gas-shift (RWGS) reaction, leading to decreased CO_2 selectivity and increased generated H_2O amount [41,55,56].

In this study, based on abundant experiments related to CO_2 selectivity such as CO conversion, catalyst preparation method, reaction temperature and K promoter, the systematical discussions are made to unveil crucial aspects in influencing CO_2 selectivity.

Fig. 8 reflects the relationship between CO_2 selectivity and CO conversion over *in situ* deposited Fe on SiO_2. Indeed, a higher CO conversion always corresponds to a higher CO_2 selectivity. When CO conversion is low, the catalyst surface has a low concentration of dissociative C* and O*. And the surface concentration of H* might be relatively higher since the adsorption and dissociation of CO molecules are competitive with that for H_2 molecules. Although the dissociative O* kinetically tends to react with CO* to form CO_2 rather than with H* to form H_2O, the higher surface H* concentration could enhance the possibility of O* which reacts with H* to reduce CO_2 formation. When CO conversion further increases, the surface H* concentration becomes lower correspondingly. Thus, the dissociative O* mainly shifts to react with CO* to form CO_2. Once CO conversion is over 60%, it seems that CO_2 selectivity does not further elevate and keeps at relatively high value (41–42%). It should be noted that over *in situ* deposited Fe on SiO_2, χ-Fe_5C_2 is major Fe phase with the content of 80–90% and Fe_3O_4 is absent regardless of the SiO_2 particle size (Fig. 5a-c and Table S3). Due to the absence of Fe_3O_4, it can be inferred that CO_2 selectivity reduced *via* RWGS reaction on Fe_3O_4 phase is negligible. In this case, the fact that lower CO conversion corresponds to lower CO_2 selectivity is because the supplying rate of adsorbed non-dissociative CO* is insufficient to react with O* to form CO_2. Thus, surface O* tends to react with H* to form H_2O in spite of a higher energy barrier.

As shown in Fig. 10, different from above case in spite of a low CO conversion, the same CO conversion at 38.3% over *in situ* deposited Fe on SiO_2 and the conventional impregnated 5.3Fe/SiO_2 does not display the same CO_2 selectivity but gives different CO_2 selectivity at 33.1% and 27.6%, respectively. It might be mainly ascribed to the notably different Fe phase composition. The spent 5.3Fe/SiO_2 consists of main Fe_3O_4 (47%) and minor χ-Fe_5C_2 (14.5%) (Fig. 5d and Table S3) due to the strong metal-support interaction. Therefore, it is reasonable to deduce that lower CO_2 selectivity over 5.3Fe/SiO_2 results from RWGS reaction on Fe_3O_4 which consumes formed CO_2 on χ-Fe_5C_2. Indeed, H_2O_{exp} increases from 3.74 to 4.24 g (Table 3). This hypothesis is in a good agreement with our previous study on CO_2 hydrogenation over Fe_3O_4 catalysts [22]. The reaction mechanisms are considered as the first step is the CO_2 reacting with H_2 to form CO and H_2O *via* the RWGS reaction over Fe_3O_4 and the second step is the formed CO reacting with H_2 to produce hydrocarbons *via* the FTS reaction over χ-Fe_5C_2.

Interestingly, as shown in Fig. 11, the lower CO conversion over precipitated 99.2Fe0.8 K-320 °C unexpectedly results in obviously higher CO_2 selectivity compared to that for 100Fe-320 °C without K promoter. In spite of significantly varied CO conversion over 99.2Fe0.8 K-320 °C, CO_2 selectivity is always stable with TOS and keeps at a high level of 42–43%. From Mössbauer spectra in Fig. 5g, the spent 99.2Fe0.8 K-320 °C consists of 100% χ-Fe_5C_2 due to deep carburization. Although CO conversion over 99.2Fe0.8 K-320 °C is lower, its surface is highly rich in active C* species due to the high reaction temperature and the K-promoted CO dissociation. Therefore, the surface H* concentration is very low, which leads to minor reaction pathway of O* towards its hydrogenation to form H_2O on χ-Fe_5C_2 and high CO_2 selectivity. Meanwhile, the disappearance of Fe_3O_4 phase suppresses the RWGS reaction to produce H_2O. As a result, H_2O_{exp} is minimum at 0.88 g.

From above discussions, the factor determining CO_2 selectivity is very complicated, which depends on CO conversion, Fe phase composition, reaction conditions, and the presence of K promoter. As correlating CO_2 selectivity with these factors, the following basic concept related to the formation of CO_2 could be established. Generally, Fe catalyst consists of Fe_3O_4 and χ-Fe_5C_2. RWGS reaction proceeds on Fe_3O_4, which could reduce CO_2 selectivity and increase the generated amount of H_2O. However, over χ-Fe_5C_2, CO dissociates into surface C* and O*. Surface O* could react with CO* to form CO_2 *via* the Boudouard reaction or with H* to form H_2O *via* the direct hydrogenation. Despite that DFT calculations show that surface O* is favorable to react with CO* to form CO_2 due to a lower energy barrier, surface O* reacting with CO* is competing with its reaction with H*. Indeed, the reaction of O* towards which pathway also significantly depends on the concentration of C* and H* on Fe catalyst.

4. Conclusions

A new process, *in situ* deposition method, has been applied to iron-based FTS reaction, which could clearly unveil the Fe phase evolution behavior by comparing the FTS reaction over the prepared Fe catalysts with the conventional impregnation method.

For *in situ* deposition method, Fe(CO)*x*-containing syngas was fed into a fixed-bed reactor under FTS reaction conditions. The fabrication of Fe catalyst takes place in parallel with FTS reaction. In this process, Fe(CO)*x* was firstly decomposed to metallic Fe on SiO_2 support. The metallic Fe could be rapidly and efficiently carburized into active χ-Fe_5C_2 as its content is as high as 80–90% in spent *in situ* deposited Fe catalysts. In contrast to *in situ* deposition method, the 5.3Fe/SiO_2 prepared by the conventional impregnation method requires pre-reduction treatment prior to the FTS reaction. Due to the strong Fe oxide-support interaction, Fe oxides including iron silicate and Fe_2O_3 could not be efficiently reduced into desired metallic Fe but terminating as FeO. The reduced FeO is not favorable to carburize into χ-Fe_5C_2 but tends to oxidize into Fe_3O_4 during the FTS reaction. As a result, the spent 5.3Fe/SiO_2 consists of the χ-Fe_5C_2 content of 14.5% and the Fe_3O_4 content of 47%. In addition, the iron silicate $Fe_2(SiO_3)_3$ could in part evolve into Fe_3O_4 in syngas with a long reaction time (60 h) as its content decreases from 39.9% in reduced catalyst to 24.5% in spent catalyst.

Besides, the factors determining CO_2 selectivity are systematically discussed based on abundant experiments. Results indicate that CO_2 selectivity depends on CO conversion, Fe phase composition, reaction conditions, and the presence of K promoter. In general, the dissociative O* from CO is kinetically favorable to react with adsorbed CO* to form CO_2 *via* Boudouard mechanism on χ-Fe_5C_2 and its hydrogenation into H_2O is hindered due to a higher energy barrier. Under the real reaction conditions, the dissociative O* reacting with CO* or H* significantly depends on the surface H*/C* ratio on χ-Fe_5C_2. When the H*/C* ratio is enhanced, the reaction of dissociative O* shifts towards hydrogenation to H_2O in spite of a higher energy barrier. Additionally, CO_2 selectivity could be reduced *via* RWGS reaction on Fe_3O_4 with H_2O generation.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.apcata.2019.01.001.

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